

TRUST//INTERCEPT

INTERCEPT THE DECISION.

Aegis Signal Lab

HackAI Malaysia 2026 — Track 03: Scam Defence & Awareness

1. Problem

Scams manipulate people into making fast decisions: click a link, pay money, disclose an OTP, install software, or trust an impersonator. A binary scam score does not explain the manipulation or give the user a safe path forward.

2. Solution

TRUST//INTERCEPT is an agentic decision-defence system. It converts suspicious text, URLs, QR codes, screenshots, documents and voice evidence into a common evidence model, then uses Hunter, Skeptic and Verifier roles to challenge the case before an Evidence Arbiter forms a conclusion.

3. Agentic Workflow

INTAKE → NORMALISE → ROUTE → HUNT → SKEPTIC → VERIFY → ARBITRATE → INTERCEPT → HUMAN APPROVAL → VERIFY / PAUSE / REPORT → LEARN. Suspicious content remains untrusted data and cannot redefine agent instructions or trigger consequential actions.

4. Differentiation

The distinctive combination is adversarial evidence review, manipulation mapping, counterfactual verification, a human approval gate, replayable provenance and adaptive scam-immunity training. The goal is not merely to classify a scam, but to defend the user's next decision.

5. Safety and Privacy

PII is redacted before hosted reasoning in the live demo. The competition demo uses a real hosted LLM provider through a server-side API key; provider keys are never placed in the browser. Deterministic local analysis remains available as a recovery mode. Suspicious destinations are not opened in the user's browser, and consequential actions require human approval.

6. Deployment

The competition deployment uses a Vercel-hosted React/Vite frontend connected over HTTPS to a separately hosted FastAPI backend. The Vercel build receives only the public `VITE\_API\_BASE\_URL`; LLM and reputation API keys remain server-side.

7. Evaluation

The built-in Evaluation Lab covers scam, legitimate, borderline and adversarial cases and reports accuracy, precision, recall, F1 and false-positive rate. The maintained benchmark currently passes with 100% accuracy, 100% precision, 100% recall, 100% F1 and 0% false-positive rate. These figures describe the maintained benchmark only and are not real-world detection guarantees.

8. Demonstration

The primary three-minute scenario is a parcel-delivery scam. The Vercel frontend sends the case to the live FastAPI backend and hosted provider. The user sees the requested action and manipulation chain, watches Hunter/Skeptic/Verifier challenge the evidence, receives an independent verification path, approves a safe next step, then opens the Evidence Passport and Case Replay. The final step becomes a short immunity challenge.

9. Expected Impact

TRUST//INTERCEPT aims to reduce impulsive scam decisions by making manipulation visible, verification independent and consequential action human-controlled. It is designed for everyday users while remaining auditable enough for security teams, educators and researchers.

10. Limitations

The system cannot guarantee scam detection. Reputation, domain and external intelligence may be unavailable. Voice analysis does not prove a deepfake. Independent verification may remain inconclusive. The benchmark is limited and should not be interpreted as real-world perfect detection. The safest output can therefore be PAUSE rather than a forced verdict.